

States = { Rested, Tired, Exhausted }

Actions = { study, Hangout, sleep }

$$P(\text{study}) = \begin{bmatrix} 0.2 & 0.8 & 0 \\ 0 & 1 & 0.9 \\ 0 & 0 & 1 \end{bmatrix}$$

* Rows represent current state.
* Columns represent next state.

eg $P(\text{tired} / \text{Rested}, \text{study}) = 0.8$

$$P(\text{Hangout}) = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.3 & 0.7 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P(\text{sleep}) = \begin{bmatrix} 1 & 0 & 0 \\ 0.8 & 0.2 & 0 \\ 0.6 & 0.4 & 0 \end{bmatrix}$$

$$R(s, a) = \begin{array}{ccccc} & & a_1 & a_2 & a_3 \\ \begin{array}{c} s_1 \\ s_2 \\ s_3 \end{array} & \begin{array}{c} 10 \\ 5 \\ -5 \end{array} & \begin{array}{c} 8 \\ 4 \\ -2 \end{array} & \begin{array}{c} 0 \\ 2 \\ 4 \end{array} \end{array}$$

$$V(s) = \max_a [R(s,a) + \gamma \sum_{s'} P(s'/s,a) V(s')]$$

~~for~~ for action = a_1 :

$$V(s_1) = 10 + (0.9)[0] = 10$$

for a_2 :

$$V(s_1) = 8$$

$$V(s_1) = 0 \text{ for } a_3$$

$$\Rightarrow V(s_1) = \max[10, 8, 0] = 10$$

lets assume after this state we transitioned to state (tired).

for action a_1 :

$$V(s_2) = 5 + 0.9[0] = 5$$

for a_2 :

$$V(s_2) = 4 + 0.9[0] = 4$$

for a_3 :

$$\begin{aligned} V(s_2) &= 2 + 0.9[(0.8)(10)] \\ &= 2 + 7.2 = 9.2 \end{aligned}$$

$$\Rightarrow V(s_2) = \max[5, 4, 9.2] = 9.2$$